Nestora: Furniture & Appliance Rental Platform
IEEE Style Project Report

Abstract

Nestora is a full-stack web application developed to provide a flexible furniture and appliance rental platform for students, professionals, and families. The system allows users to browse products, authenticate, place rental orders, and manage profiles through a responsive web interface backed by Node.js, Express.js, and MongoDB.

This section discusses the design, implementation, and relevance of the Nestora platform in greater detail. The project demonstrates full-stack web development principles, modular architecture, RESTful communication, database integration, and practical deployment considerations for an academic major project.

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Keywords

Furniture Rental, Node.js, Express.js, MongoDB, REST API, Web Application

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1. Introduction

Nestora addresses the growing demand for affordable rental solutions instead of ownership. It enables users to rent essential household items for flexible durations.

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2. Problem Statement

Buying furniture is expensive for temporary residents. Existing solutions often lack affordability and ease of use.

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3. Objectives

Provide an intuitive rental platform, secure authentication, product catalog, order management, and admin operations.

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4. Existing System

Traditional rental services rely on manual processes with limited online functionality.

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5. Proposed System

A centralized online rental platform with authentication, browsing, cart, checkout, and backend management.

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6. Technology Stack

Frontend: HTML, CSS, JavaScript. Backend: Node.js and Express.js. Database: MongoDB using Mongoose.

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7. System Architecture

Client communicates with Express REST APIs, which process business logic and interact with MongoDB.

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8. Database Design

Core collections include Users, Products, Rentals, Maintenance Requests, and Business data.

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9. Modules

Authentication, Product Management, Cart, Checkout, Profile, Admin Dashboard, Rental Management.

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10. Workflow

User registers, logs in, browses products, adds items to cart, checks out, and views rental history.

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11. Implementation

The application uses RESTful APIs, JSON data exchange, server-side routing, and persistent MongoDB storage.

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12. Testing

Functional testing verified authentication, product browsing, cart operations, checkout flow, and backend APIs.

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13. Results

The system successfully demonstrates an end-to-end furniture rental workflow suitable for academic purposes.

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14. Advantages

Affordable access to furniture, user-friendly interface, centralized management, and scalable architecture.

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15. Security

Passwords are securely hashed using crypto.scrypt, with environment-variable based configuration.

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16. Future Scope

Payment gateway integration, AI recommendations, mobile application, notifications, and analytics.

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17. Conclusion

Nestora provides a practical, modern rental platform with clean architecture and a strong academic implementation.

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